CLINICAL · NECK, THYROID & SALIVARY

Thyroid nodules, cancer and goitre

Thyroid nodules are everywhere: one in twenty adults can feel one, and ultrasound finds them in half of older people. The task is to find the small fraction that are cancers without operating on everyone. The tools are history, ultrasound pattern, and ultrasound-guided FNA reported in Bethesda categories. Most thyroid cancers are papillary and have an excellent outlook. The surgeon's measure is two working recurrent laryngeal nerves and four living parathyroids at the end of the operation.

AT A GLANCE

- Ultrasound and FNA guide nodule care.
- Bethesda sets the next step.
- Check TSH and the vocal folds.
- Watch nerve, calcium and bleeding after surgery.
- U1–U5 and Thy1–5 grading; NIFTP is benign; low-dose iodine (HiLo) for low-risk disease.

1 CHAPTER ONE Applied anatomy

The gland and the nerves around it.

1.1 Embryology

The medial anlage starts at the *foramen caecum* of the tongue at about four weeks. It descends in front of the hyoid and larynx along the *thyroglossal tract* and reaches the root of the neck by seven weeks. Arrested descent leaves a *lingual thyroid*, often the only thyroid tissue the patient has; the lowest part of the tract persists as the *pyramidal lobe* in up to 40%; and remnants of the tract form *thyroglossal cysts*.

The lateral anlagen of the fourth and fifth pouches and the *ultimobranchial bodies* join the descending gland. The ultimobranchial bodies bring the neural-crest *C cells*, which concentrate in the upper and middle thirds of the lobes, and that is where medullary cancers arise.

1.2 The gland

The gland has two lobes joined by an isthmus over the second to fourth tracheal rings. It weighs 15 to 25 g, and each lobe measures about 5 × 3 × 2 cm. Its *true capsule* sends septa into the gland, while its *false capsule*, the perithyroidal sheath, is formed by the visceral (pretracheal) layer of the deep cervical fascia.

Two suspensory ligaments hold it. The anterior ligament runs from the isthmus to the trachea, and the posterior ligament, the **ligament of Berry**, binds the back of each lobe to the cricoid and first tracheal rings, which is why the gland rises on swallowing. The recurrent laryngeal nerve runs deep to or through Berry's ligament beside small branches of the inferior thyroid artery. This is the commonest site of nerve injury, and thyroid tissue left here explains bed uptake and measurable thyroglobulin after a "total" thyroidectomy.

The **superior thyroid artery** is the first branch of the external carotid. The **inferior thyroid artery** arises from the thyrocervical trunk, passes behind the carotid sheath and supplies the parathyroids, and a *thyroidea ima* sometimes arises from the brachiocephalic trunk or the arch. The superior and middle veins drain to the internal jugular; the middle vein has no companion artery and is the first vessel divided at surgery; and the inferior veins form a plexus draining to the brachiocephalic veins. Lymph passes first to the central compartment (level VI), to the prelaryngeal (*Delphian*), pretracheal and paratracheal nodes, and then to the superior mediastinum (level VII) and the lateral neck (levels II to IV, sometimes V).

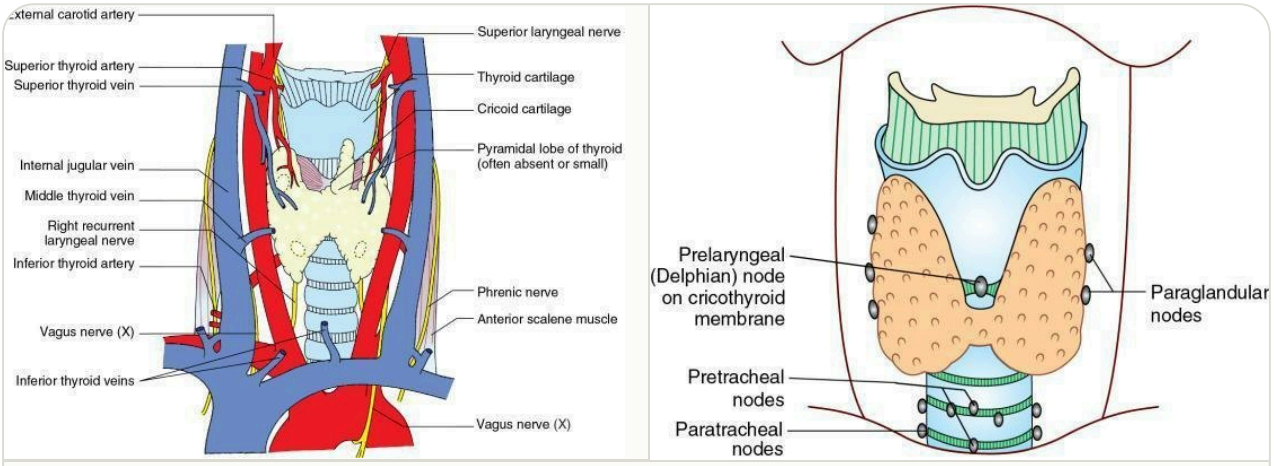


FIGURE 1 First, the thyroid and its relations. Second, lymphatic drainage of the thyroid: Delphian, pretracheal and paratracheal nodes, the jugular chain and the mediastinum.

Dhingra, figs 66.1 and 66.2

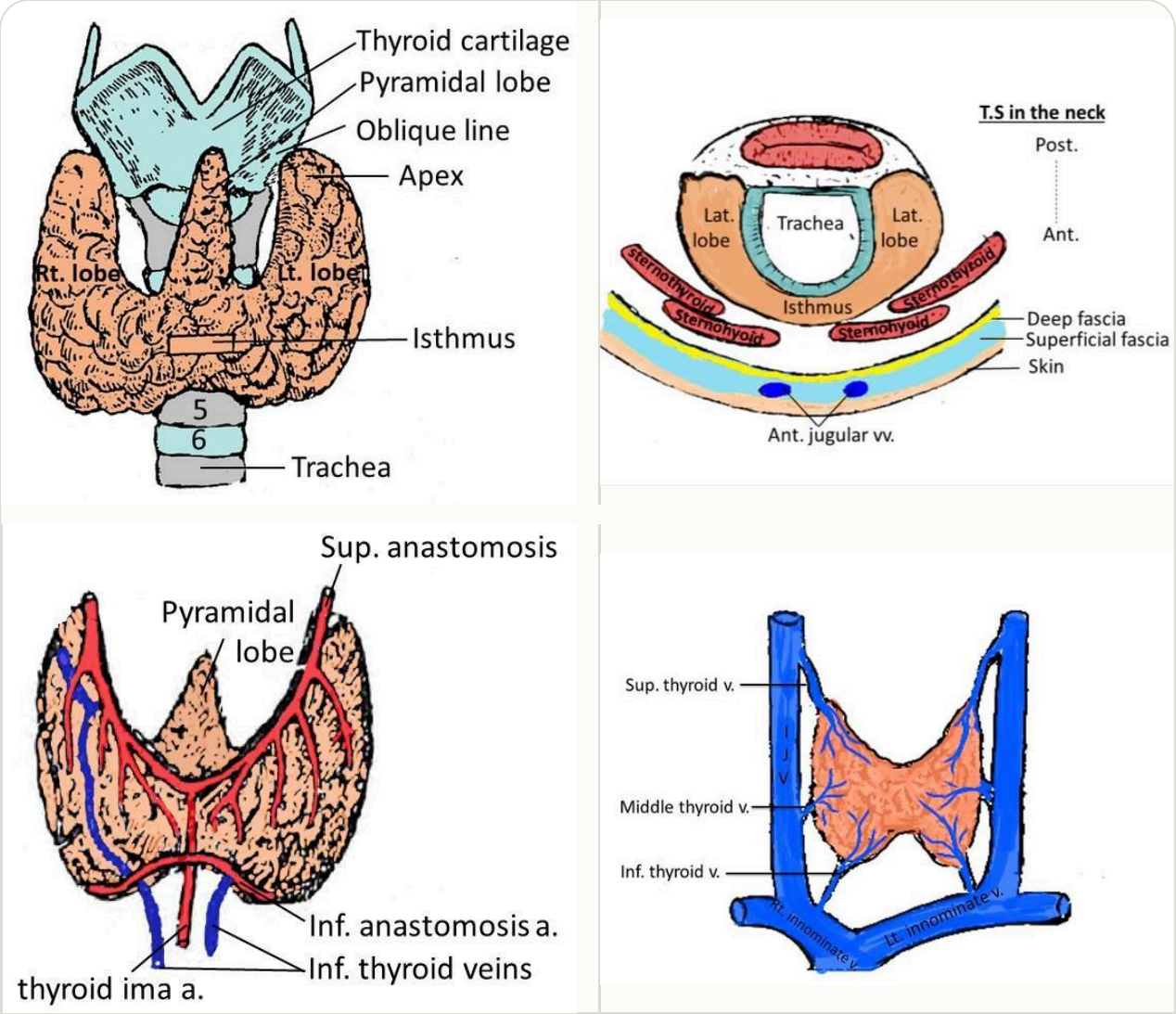


FIGURE 2 The thyroid from the front, in cross-section with its capsules, and its arteries and veins.

Anatomy atlas · Doss p. 124, Doss p. 125, Doss p. 128

1.3 Nerves at risk

The **recurrent laryngeal nerve** supplies all the intrinsic laryngeal muscles except cricothyroid and carries sensation below the folds. The *right* nerve hooks round the subclavian artery and runs obliquely from lateral to medial; the *left* hooks round the aortic arch and runs more vertically in the tracheo-oesophageal groove. Each crosses the branches of the inferior thyroid artery, superficial to them, deep to them or between them, branches before entering the larynx in about a third of people, and enters behind the inferior cornu of the thyroid cartilage. It is found in the *recurrent nerve triangle*, bounded by the trachea medially, the retracted strap muscles laterally and the lower pole above, or close to the *tubercle of Zuckerkandl*.

A **non-recurrent laryngeal nerve** occurs in about 0.5 to 1% of people on the right and almost never on the left. It runs directly from the vagus to the larynx and accompanies an aberrant retro-oesophageal right subclavian artery, the cause of *dysphagia lusoria*.

The **external branch of the superior laryngeal nerve** supplies cricothyroid, which tenses the cord for high pitch. It runs on the inferior constrictor close to the superior pole vessels and crosses them within 1 cm of the pole in about a fifth of people, so the superior vessels are ligated individually on the capsule with the gland pulled down. The **ansa cervicalis** supplies the strap muscles from below, so the straps are divided high when they must be divided.

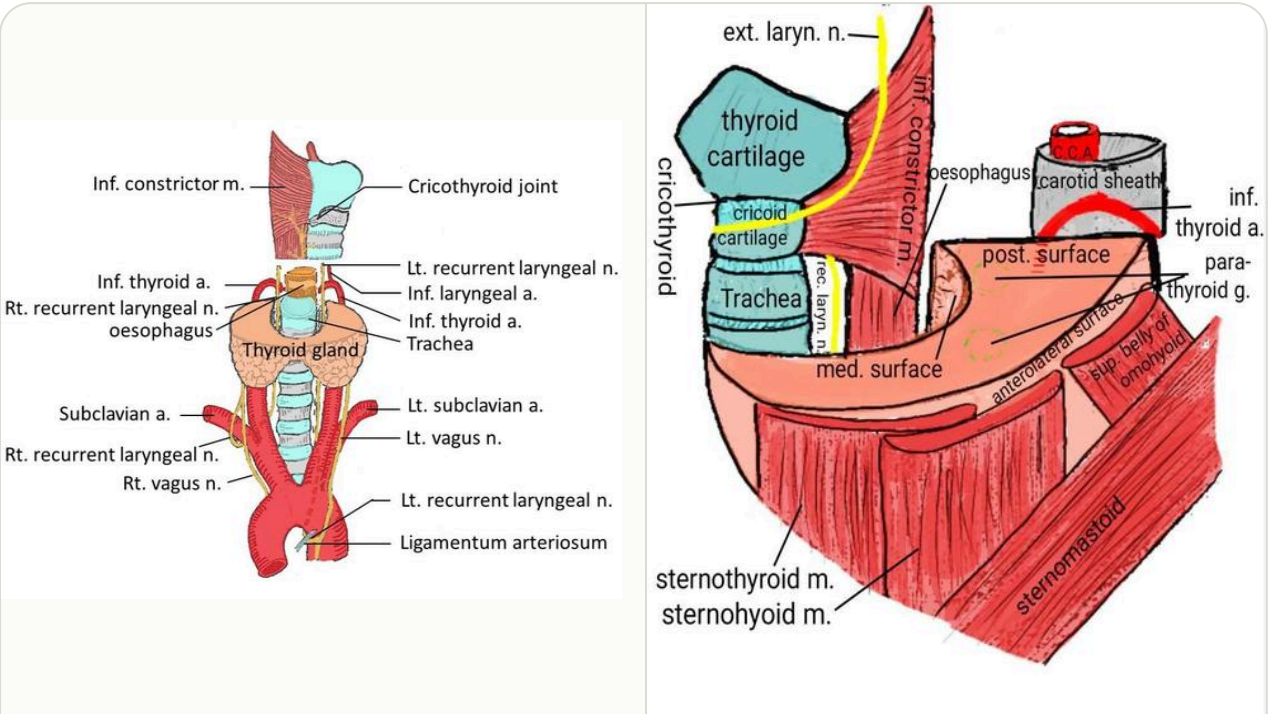


FIGURE 3 The recurrent laryngeal nerves and inferior thyroid arteries, and the relations of the lateral lobe.

Anatomy atlas · Doss p. 154, Doss p. 126

1.4 Lingual thyroid

A lingual thyroid occurs in about 1 in 3000 to 4000 patients with thyroid disease, mostly women. It is a smooth, vascular mass at the tongue base that may cause dysphagia, dysphonia, airway obstruction or bleeding, particularly at puberty or in pregnancy. Before any treatment, a scintiscan and neck ultrasound establish whether any other thyroid tissue exists. Thyroxine shrinks it and radioiodine can

ablate it; obstruction is relieved by excision through a transoral or suprahyoid transpharyngeal approach, followed by lifelong replacement if it was the only thyroid.

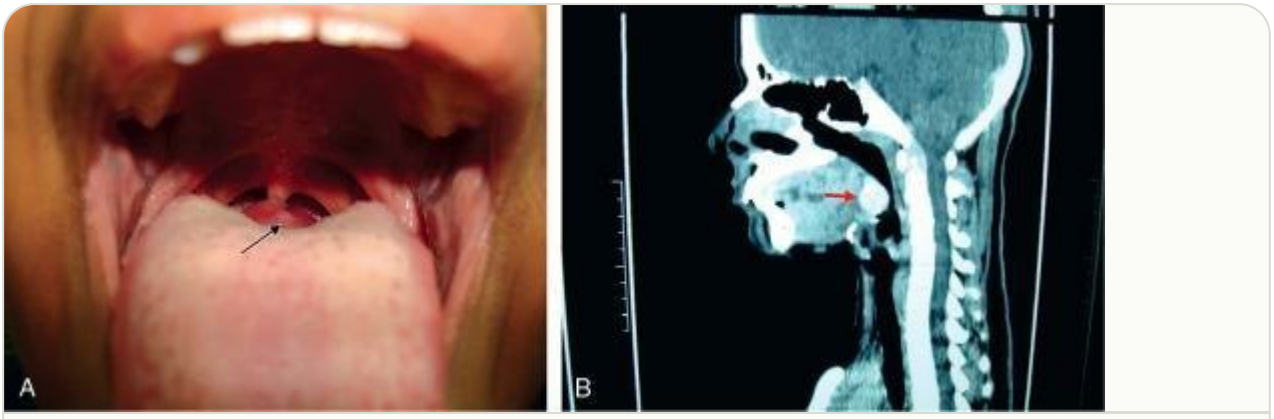


FIGURE 4 Lingual thyroid at the tongue base, with its sagittal CT.
Dhingra, fig. 66.3

2 CHAPTER TWO Physiology and function tests

Hormones and the axis.

2.1 Hormone synthesis

Follicular cells make thyroglobulin, thyroxine (T4) and tri-iodothyronine (T3), and the *parafollicular C cells* make calcitonin. Synthesis proceeds in steps:

1. Iodide is actively trapped.
2. *Thyroid peroxidase* oxidises it and attaches it to tyrosine residues on thyroglobulin, forming MIT and DIT.
3. Thyroid peroxidase couples them: DIT + DIT forms T4, and MIT + DIT forms T3.
4. The hormones are stored as colloid.
5. Colloid is taken back into the cell, thyroglobulin is cleaved, and the hormones are released.
6. Uncoupled MIT and DIT are deiodinated so that their iodine is recycled.

The gland releases mostly T4, with a half-life of about a week, and most T3, several times more potent and with a half-life of about a day, is made by peripheral deiodination. More than 99% of circulating hormone is protein-bound.

Hypothalamic TRH stimulates pituitary TSH, TSH drives hormone synthesis, growth and vascularity of the gland, and T4 and T3 feed back negatively.

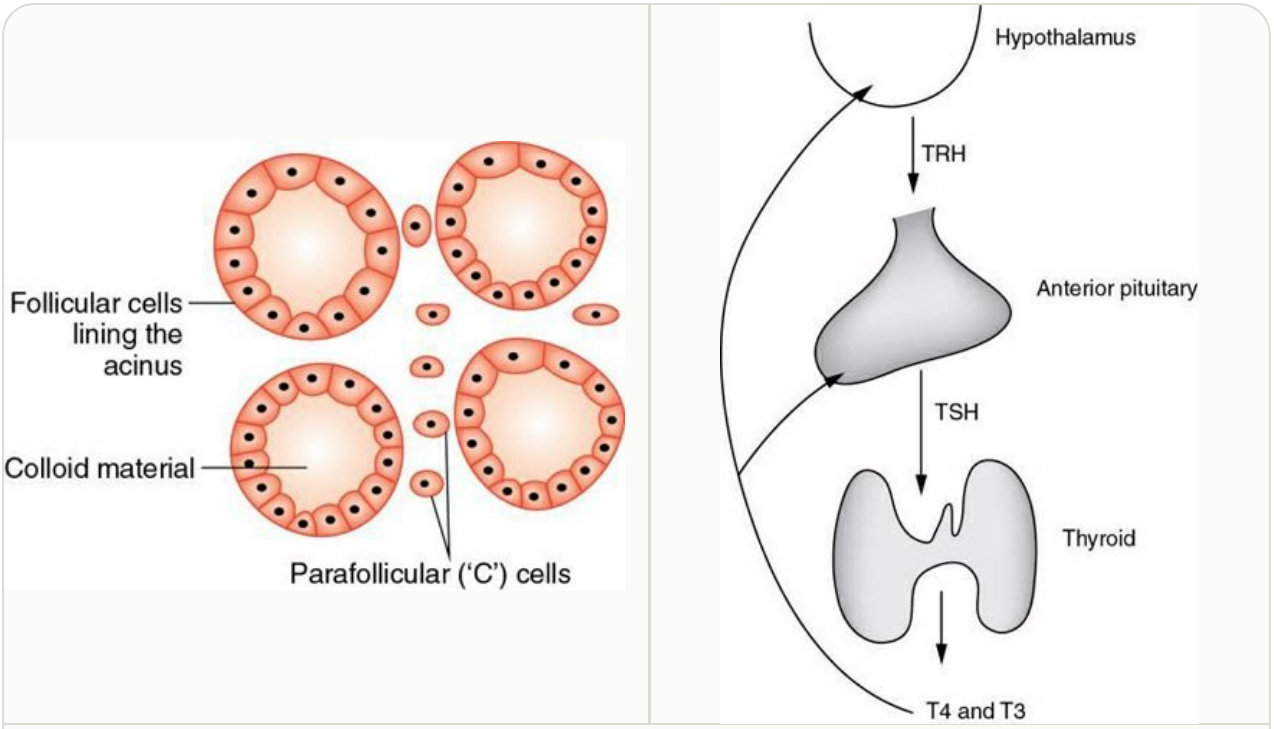


FIGURE 5 First, thyroid follicles with follicular and parafollicular C cells. Second, the hypothalamic–pituitary–thyroid axis.
Dhingra, figs 66.4 and 66.5

2.2 Tests and terms

TSH is the single best screening test: high in primary hypothyroidism and suppressed in hyperthyroidism. A high TSH with normal T4 is subclinical hypothyroidism, and a low TSH with normal T4 and T3 subclinical hyperthyroidism. **Free** T4 and T3 are measured because total levels vary with thyroid-binding globulin, which rises in pregnancy and with the pill.

TPO (antimicrosomal) antibodies are present in almost all Hashimoto disease and most Graves disease; **thyroglobulin antibodies** occur in autoimmune thyroiditis and interfere with thyroglobulin assays; and stimulating **TSH-receptor antibodies** cause Graves disease. **Thyroglobulin** is made only by thyroid tissue; it is useless for diagnosing a nodule, but after total thyroidectomy and ablation it should be undetectable, so a rising level means recurrence. **Calcitonin and CEA** are the markers of medullary carcinoma.

Ultrasound is the essential imaging study. **Scintigraphy** with technetium or I-123 shows hot and cold nodules, and **CT**, with non-iodinated contrast if radioiodine may follow, shows retrosternal extension and invasion.

3 CHAPTER THREE Benign disease

Function and inflammation.

3.1 Hypothyroidism

Iodine deficiency is the commonest cause of hypothyroidism worldwide and *Hashimoto disease* the commonest where iodine is sufficient. Thyroidectomy and radioiodine, neck irradiation (almost half of patients after laryngeal or pharyngeal radiotherapy, so TSH is checked routinely), drugs such as lithium, amiodarone and antithyroid drugs, inborn defects, and rare pituitary failure are the other causes.

The patient is tired, intolerant of cold, gains weight, has dry coarse skin and hair, constipation, menorrhagia, slow thinking, bradycardia and non-pitting *myxoedema*. In ENT terms the voice becomes hoarse and low from oedematous cords, the tongue enlarges, and sensorineural hearing loss and serous otitis occur. Congenital hypothyroidism (cretinism), with stunted growth, intellectual disability and deafness, is prevented by neonatal screening. Treatment is levothyroxine, started low in the elderly and in heart disease. Anaesthesia and sedatives can precipitate *myxoedema coma* in untreated patients.

3.2 Hyperthyroidism

Hyperthyroidism causes weight loss despite a good appetite, heat intolerance, sweating, tremor, palpitations and atrial fibrillation, diarrhoea, anxiety, proximal myopathy and oligomenorrhoea. Graves disease is the commonest cause and toxic multinodular goitre the commonest in the elderly; toxic adenoma, transient thyroiditis, iodine excess (the *Jod–Basedow phenomenon* after contrast in a nodular goitre) and exogenous hormone follow, with rare TSH-secreting pituitary tumours, struma ovarii and functioning cancers.

CAUSES OF HYPERTHYROIDISM AND THEIR SCANS

CAUSE	KEY FEATURES	ISOTOPE SCAN
Graves disease	Diffuse goitre, bruit, eye disease, pretibial myxoedema, TSH-receptor antibodies	Diffuse high uptake
Toxic multinodular goitre	Elderly, nodular goitre, no eye signs	Patchy hot areas, suppressed rest
Toxic adenoma	Single nodule, usually over 2.5 cm, often T3 toxicosis	One hot nodule, rest suppressed
Subacute (de Quervain) thyroiditis	Painful tender gland after a viral illness, raised ESR	Very low uptake
Silent and postpartum thyroiditis	Painless, transient	Very low uptake

3.3 Graves disease

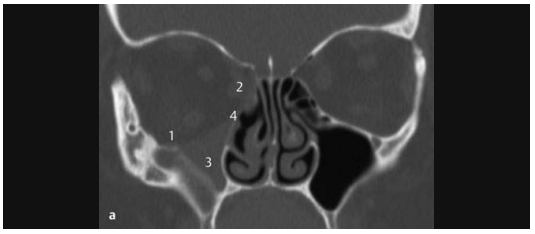
Graves disease is an autoimmune disease in which stimulating TSH-receptor antibodies drive the gland. Women are affected five to ten times more often than men, usually in the third and fourth decades. It produces a diffuse vascular goitre with a bruit, thyrotoxicosis, *ophthalmopathy* with lid retraction, proptosis, diplopia and optic neuropathy, and dermopathy as pretibial myxoedema.

Antithyroid drugs are first line: carbimazole or methimazole, or propylthiouracil in the first trimester despite its risk of liver failure. All carry a risk of agranulocytosis, so the patient is told to report a sore throat, and relapse after stopping is common. **Beta-blockers** control symptoms. **Radioiodine** is definitive but usually leads to hypothyroidism; it is avoided in pregnancy and breastfeeding, conception is deferred for at least six months, and it can worsen eye disease, especially in smokers. **Total thyroidectomy** is chosen for drug failure or side effects, a large goitre or nodules, active eye disease, pregnancy planning, or the patient's preference.

Before surgery the patient must be made euthyroid on drugs to avoid *thyroid storm*, and Lugol's iodine for about ten days reduces vascularity through the Wolff–Chaikoff effect. Severe orbitopathy is treated by endoscopic transnasal orbital decompression.

IMAGING IN THE APP 1 case

Endoscopic orbital decompression in Graves disease.



Orbital Decompression in Graves Disease

After surgery · 2 images

3.4 Thyroiditis

Hashimoto (chronic lymphocytic) thyroiditis is the commonest thyroid disease where iodine is sufficient, mostly in women. It produces a painless, firm, rubbery, symmetrical goitre with raised TPO antibodies, and the patient becomes hypothyroid over time; histology shows lymphoid follicles and Hürthle cell change. It is treated with thyroxine, but a discrete nodule still needs FNA, and a rapidly growing mass raises the possibility of *lymphoma*.

Subacute granulomatous (de Quervain) thyroiditis is the commonest painful thyroid. It follows a viral infection, with neck pain radiating to the jaw and ear, fever, a tender gland and a high ESR, and passes through a transient toxic phase with very low isotope uptake, a hypothyroid phase and recovery. It is treated with NSAIDs, steroids and beta-blockers. **Silent and postpartum thyroiditis** are painless and transient but sometimes leave permanent hypothyroidism.

Acute suppurative thyroiditis is rare: a bacterial infection with a tender red swelling and sometimes an abscess. In children it is usually *left-sided* and comes from a *pyriform sinus fistula*, a third or fourth branchial remnant; after antibiotics and drainage, the fistula is sought with barium swallow, CT or endoscopy and excised or its opening cauterised. **Riedel thyroiditis** is very rare: fibrosis spreading out of the gland, part of the IgG4-related spectrum, making it woody, fixed and non-tender, compressing the trachea and paralysing the recurrent nerve, and mimicking

anaplastic carcinoma or lymphoma. It is biopsied, often by isthmusectomy, which also relieves tracheal compression, and treated with steroids or tamoxifen.

3.5 Goitre

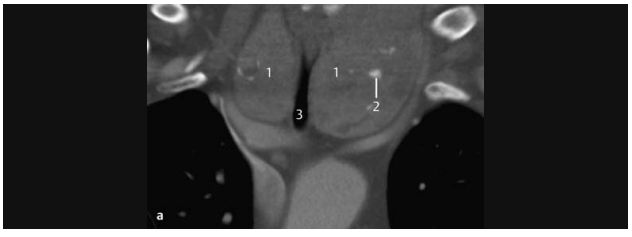
A goitre is an enlarged gland, diffuse or nodular. It results from endemic iodine deficiency, goitrogens, inherited defects of hormone synthesis (*Pendred syndrome*, with sensorineural deafness and an enlarged vestibular aqueduct), puberty and pregnancy, autoimmunity and nodular hyperplasia. Endemic goitre is prevented by iodised salt.

The **multinodular goitre** evolves slowly and may become toxic. A sudden painful enlargement means haemorrhage into a nodule, and about 5 to 10% harbour a carcinoma, usually a small papillary one. Pressure causes cough, choking, positional or nocturnal breathlessness, stridor and dysphagia, and superior vena caval obstruction gives a *positive Pemberton sign*: raising the arms brings on facial plethora, distended neck veins and breathlessness.

Assessment is by TSH, ultrasound with FNA of suspicious or dominant nodules, laryngoscopy, and CT for retrosternal extension and tracheal compression, with flow–volume loops if needed. Surgery is indicated for compression, substernal extension that can no longer be examined or sampled, suspicion of cancer, toxicity and cosmesis. A retrosternal goitre is almost always removed through the neck; sternotomy is rarely needed, mainly for a primary mediastinal goitre or one extending below the aortic arch.

IMAGING IN THE APP 1 case

Goitre on imaging, and the normal thyroid.



Hypertrophy of the Thyroid Gland, Thyroid Goiter

Infection & inflammation · 11 images

OPERATIVE GUIDES IN THEATRE 1 operation

Massive and substernal goitre.

Massive and Substernal Goiter
Substernal goitre

4 CHAPTER FOUR The thyroid nodule

Finding the cancer.

4.1 Causes

Most nodules are benign: a colloid (adenomatous) nodule in a nodular goitre, a follicular or Hürthle cell adenoma, a cyst, focal thyroiditis or a regenerative nodule, or a toxic adenoma. Carcinoma accounts for about 5 to 15%, and metastasis and lymphoma are rare. Some masses only look thyroid: a node, a parathyroid cyst, a thyroglossal cyst or a laryngocele.

CLINICAL FEATURES RAISING CONCERN FOR CANCER

LOWER CONCERN	HIGHER CONCERN
Long-standing, stable nodule	Age under 20 to 30 or over 60 years; male
Multinodular gland without a dominant nodule	Childhood neck irradiation; family history of thyroid cancer or MEN 2
Hashimoto or toxic features	Rapid growth, pain; hard, fixed nodule; size over 4 cm
Soft, mobile	Cervical lymphadenopathy; vocal fold palsy; stridor, dysphagia

EXAMINE THE LARYNX IN EVERY THYROID PATIENT

A paralysed vocal fold can coexist with a normal voice. Preoperative laryngoscopy finds unsuspected palsy, which suggests invasive cancer and changes the surgical plan. It also documents the baseline before surgery.

4.2 Investigation

TSH comes first. If it is low, a scintiscan follows, and a hot nodule is almost never malignant and is not aspirated. If it is normal or high, the patient goes on to ultrasound and FNA.

Every patient has an ultrasound of the thyroid and neck, and the pattern sets the risk. High suspicion attaches to a solid hypoechoic nodule with microcalcifications, irregular or infiltrative margins, a taller-than-wide shape or extrathyroidal extension. A solid hypoechoic nodule with smooth margins is intermediate; an iso- or hyperechoic nodule, or a partly cystic one with an eccentric solid part, is low; a spongiform nodule is very low; and a purely cystic nodule is benign. Suspicious nodes show microcalcification, cystic change, loss of the fatty hilum, a round shape and peripheral vascularity.

Ultrasound-guided FNA is done for nodules of 1 cm or more with high or intermediate suspicion, 1.5 cm or more with low suspicion, and 2 cm or more with very low suspicion (or they are observed), but not for pure cysts. Suspicious nodes are also aspirated, with thyroglobulin measured in the needle washout. **Calcitonin** is measured when there is a family history of medullary cancer or MEN 2, and **CT or MRI** is used for retrosternal extension, invasion or bulky nodal disease.

THE BETHESDA SYSTEM FOR THYROID CYTOLOGY

CATEGORY	RISK OF CANCER	USUAL MANAGEMENT
I. Non-diagnostic	5 to 10%	Repeat ultrasound-guided FNA
II. Benign	0 to 3%	Clinical and ultrasound follow-up
III. Atypia of undetermined significance (AUS/FLUS)	10 to 30%	Repeat FNA, molecular testing, or lobectomy
IV. Follicular neoplasm (including Hürthle cell)	25 to 40%	Molecular testing or diagnostic lobectomy
V. Suspicious for malignancy	50 to 75%	Lobectomy or total thyroidectomy
VI. Malignant	97 to 99%	Surgery

WHY FNA CANNOT DIAGNOSE FOLLICULAR CARCINOMA

Follicular carcinoma is diagnosed by *capsular or vascular invasion*. That means examining the whole capsule histologically, which an aspirate cannot do. A "follicular neoplasm" therefore needs a diagnostic lobectomy, and completion thyroidectomy if invasion is found. Hürthle cell neoplasms behave the same way. Molecular panels (BRAF, RAS, RET/PTC, PAX8/PPARγ, gene-expression classifiers) help sort indeterminate cytology.

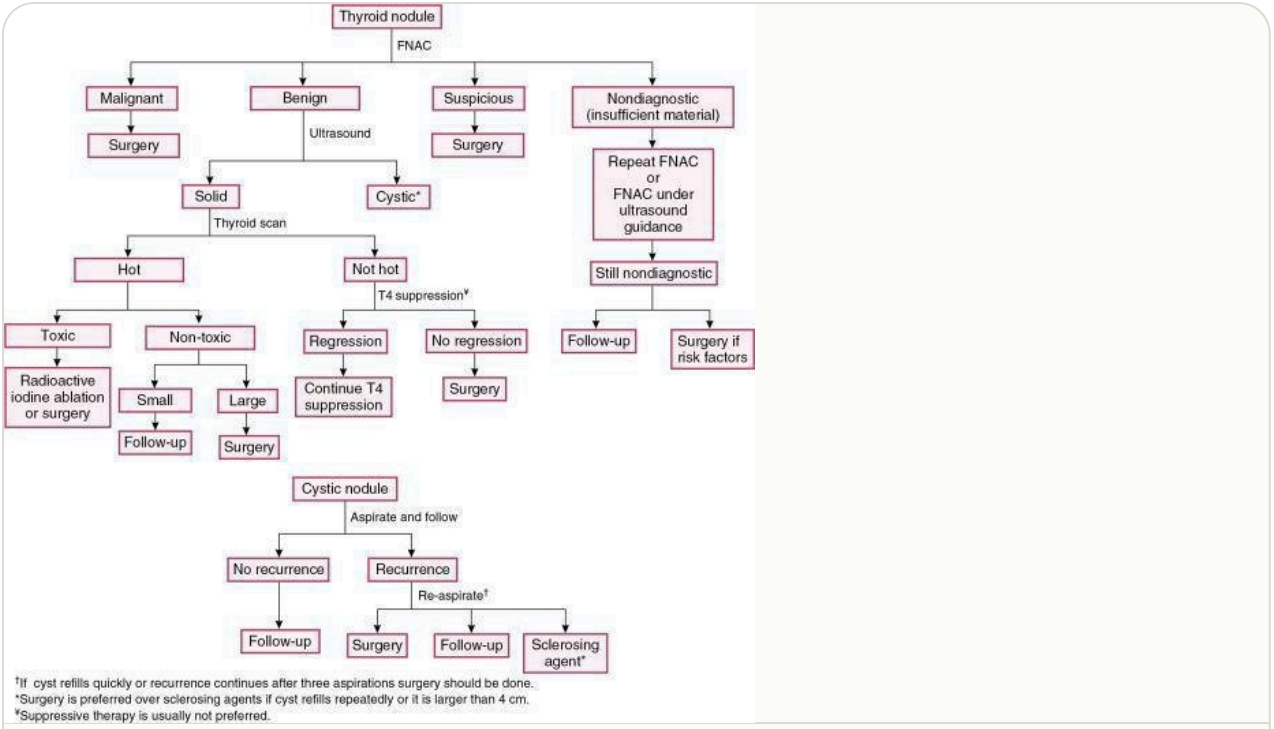


FIGURE 6 Management of a thyroid nodule.
Dhingra, fig. 66.8

4.3 Cystic nodules

About a quarter of nodules are wholly or partly cystic, and the colour of the aspirate is informative. Brown fluid is old haemorrhage into a colloid nodule. Water-clear fluid suggests a *parathyroid cyst*, confirmed by measuring PTH in the fluid. Bloody fluid that reaccumulates raises suspicion of cystic papillary carcinoma, and the solid part is aspirated under ultrasound. Cysts that recur repeatedly or exceed about 4 cm are excised, and ethanol ablation is an alternative for benign recurrent cysts.

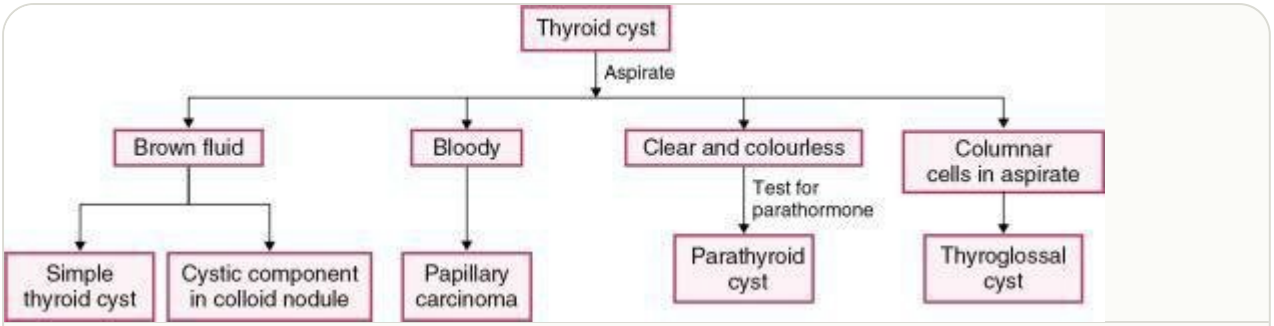


FIGURE 7 Management of a thyroid cyst according to the aspirate.
Dhingra, fig. 66.7

4.4 Benign nodules

Benign nodules are followed with examination and ultrasound, and aspirated again if they grow significantly or change pattern. TSH suppression with thyroxine is little used, because its effect is unpredictable and it harms bone and heart. A toxic adenoma is treated by lobectomy or radioiodine, and other benign nodules are removed only for compression or cosmesis, with radiofrequency or laser ablation as options for selected nodules.

5 CHAPTER FIVE Thyroid cancer

Five types.

MALIGNANT TUMOURS OF THE THYROID

TYPE	PROPORTION	CELL OF ORIGIN	SPREAD
Papillary	70 to 85%	Follicular	Lymphatic (central and lateral nodes)
Follicular, including Hürthle (oncocytic)	10 to 15%	Follicular	Blood (lung, bone)
Medullary	5%	Parafollicular C cell	Nodes early, then blood
Anaplastic	Under 5%	Follicular (dedifferentiated)	Local invasion, everywhere

TYPE	PROPORTION	CELL OF ORIGIN	SPREAD
Lymphoma	About 5%	Lymphoid (in Hashimoto)	Nodal and extranodal

5.1 Papillary carcinoma

Papillary carcinoma is the commonest type and is becoming commoner as more small tumours are detected. It usually affects women of 30 to 50, and it is also the usual childhood thyroid cancer, often with nodes and lung spread but still with an excellent prognosis. The main risk factor is *ionising radiation* in childhood, from old low-dose treatment of tonsils, acne and thymus, radiotherapy for Hodgkin disease, and Chernobyl and atomic fallout, with a latency of decades; family history, familial adenomatous polyposis, Gardner and Cowden syndromes, and adequate iodine intake also contribute.

Microscopically it forms papillae with characteristic nuclei, the ground-glass "*Orphan Annie eye*" nuclei with grooves and intranuclear inclusions, and *psammoma bodies*, and it is often multifocal. Because the diagnosis rests on nuclear features, cytology can make it. The tall cell, columnar, diffuse sclerosing and hobnail variants are aggressive, and BRAF V600E, the commonest mutation, is associated with more aggressive behaviour.

It spreads by lymphatics: clinically evident nodes are present in about 30% of adults and in many more children, and the nodal deposits may be cystic and sometimes form the presenting lump behind an occult primary. Distant spread is uncommon. Ten-year survival exceeds 90%, and prognosis worsens with older age, male sex, larger size, extrathyroidal extension and distant metastases, as captured by the *AGES* (age, grade, extent, size) and *AMES* (age, metastases, extent, size) scores. Low-risk **papillary microcarcinomas** of 1 cm or less may be watched by active surveillance or treated by lobectomy.

5.2 Follicular carcinoma

Follicular carcinoma affects an older group than papillary carcinoma, around 50, mostly women, and is commoner in iodine-deficient areas. It is usually unifocal, and the diagnosis needs capsular or vascular invasion on histology; *minimally invasive* tumours behave well and *widely invasive* ones do not. It spreads by *blood* to lung and

bone, with distant metastases in up to 15% at diagnosis, while nodal spread is uncommon. RAS mutations and the PAX8/PPARγ rearrangement are characteristic, and the prognosis is worse with age over 50, size over 4 cm and distant disease.

5.3 Hürthle cell (oncocytic) carcinoma

Hürthle cell carcinoma consists of large eosinophilic cells packed with mitochondria. It is more aggressive than follicular carcinoma, more often multifocal and with nodal and distant spread, and because it takes up *radioiodine poorly*, surgery is all-important.

5.4 Medullary carcinoma

Medullary carcinoma arises from the C cells, secretes *calcitonin* and CEA, which are used for diagnosis and follow-up, and shows amyloid stroma on histology. It is equally common in both sexes. About 75 to 80% are **sporadic**, unifocal tumours in the fifth decade. The other 20 to 25% are **familial**, caused by autosomal dominant germline *RET* mutations; they are preceded by C-cell hyperplasia, are bilateral and multifocal, and occur in MEN 2A, MEN 2B and familial medullary carcinoma, now counted as a variant of MEN 2A.

It spreads early to central and lateral nodes, then to lung, liver and bone, and its hormones cause diarrhoea and flushing. The work-up measures calcitonin and CEA, tests *every* patient for RET, excludes a *phaeochromocytoma* with plasma or urinary metanephrines before any surgery (removing it first to avoid a hypertensive crisis), and checks serum calcium for hyperparathyroidism.

Treatment is total thyroidectomy with central (level VI) neck dissection, adding a lateral dissection for nodal disease or according to the calcitonin level. Radioiodine does not work, and RET and multikinase inhibitors such as selpercatinib, vandetanib and cabozantinib treat progressive metastatic disease. RET carriers have a prophylactic thyroidectomy *within the first year* for the highest-risk (MEN 2B) mutations, *by about five years* for high-risk MEN 2A mutations, and later, guided by calcitonin, for moderate-risk mutations.

MULTIPLE ENDOCRINE NEOPLASIA

SYNDROME	GENE	COMPONENTS
MEN 1 (Wermer)	MEN1	Parathyroid hyperplasia, pituitary tumours, pancreatic islet tumours
MEN 2A (Sipple)	RET	Medullary carcinoma (nearly all), phaeochromocytoma (about 50%), hyperparathyroidism (20 to 30%), cutaneous lichen amyloidosis, Hirschsprung disease
MEN 2B	RET (M918T)	Medullary carcinoma (early, aggressive), phaeochromocytoma, mucosal neuromas of lips and tongue, marfanoid habitus, intestinal ganglioneuromatosis; no hyperparathyroidism

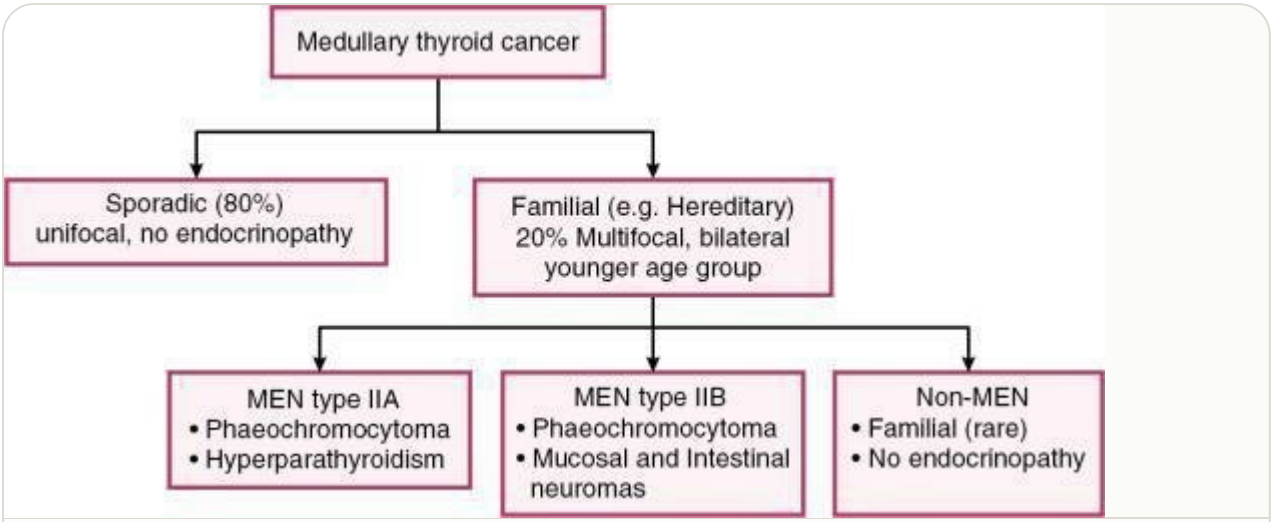


FIGURE 8 Types of medullary carcinoma: sporadic (about 80%) and familial (MEN 2A, MEN 2B, and familial non-MEN).
Dhingra, fig. 66.6

5.5 Anaplastic carcinoma

Anaplastic carcinoma is one of the most lethal cancers, with a median survival of about six months. It arises in old people, usually over 60, often by dedifferentiation of a long-standing differentiated cancer, and is always stage IV. It presents as a rapidly enlarging, hard, often painful mass fixed to its surroundings, with hoarseness, stridor, dysphagia and nodes, and half the patients have distant metastases.

A core or open biopsy must provide enough tissue to exclude *lymphoma*, which is curable. The tumour is resected only if resection can be complete; otherwise it is treated with radiotherapy and chemotherapy, and BRAF-mutant tumours respond to

dabrafenib with trametinib. Palliating the airway, including the hard decision about tracheostomy through tumour, and supportive and palliative care are central.

5.6 Lymphoma

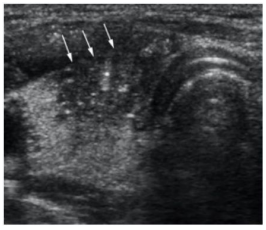
Thyroid lymphoma usually affects older women and almost always arises in *Hashimoto thyroiditis*. It is mostly a B-cell non-Hodgkin lymphoma, diffuse large B-cell or MALT type, presenting as a rapidly enlarging, painless, firm goitre with compression, often in a hypothyroid patient. It is diagnosed by core or open biopsy with immunohistochemistry and flow cytometry, and treated with chemotherapy and radiotherapy, surgery being limited to biopsy or relief of the airway. The outlook is good.

5.7 Metastases to the thyroid

Metastases reach the thyroid from the kidney, breast, lung, melanoma and head and neck squamous carcinoma.

IMAGING IN THE APP 1 case

Thyroid carcinoma on imaging.



Thyroid cancer

Tumours · 10 images

6 CHAPTER SIX Treating differentiated cancer

Surgery, iodine, suppression.

6.1 Extent of thyroidectomy

Lobectomy with isthmusectomy is enough for low-risk, unifocal, intrathyroidal tumours up to about 4 cm without nodes or extrathyroidal extension. **Total thyroidectomy** is needed for tumours over 4 cm, gross extrathyroidal extension, nodal or distant metastases, bilateral disease, previous irradiation, and whenever radioiodine is planned. **Completion thyroidectomy** follows when lobectomy histology shows a higher-risk tumour.

6.2 The neck

Clinically involved central nodes are removed by a **therapeutic central (level VI) dissection**. A **prophylactic central dissection** is debated; it is considered for T3 and T4 tumours and when lateral nodes are involved, at the cost of more hypoparathyroidism. Lateral nodes proven on FNA are treated by a compartmental selective neck dissection of levels II to IV, with V as needed, and never by "berry picking".

OPERATIVE GUIDES IN THEATRE 1 operation

Central compartment neck dissection.

Central Compartment Neck Dissection
Central dissection

6.3 Invasive disease

A functioning recurrent nerve with tumour on its surface is shaved and kept, while a paralysed or encased nerve is resected. Involved strap muscles are removed en bloc, and invasion of the trachea, larynx or oesophagus is treated by shave or full-thickness window or sleeve resection according to its depth.

OPERATIVE GUIDES IN THEATRE 1 operation

Invasive thyroid cancer.

Invasive Thyroid Cancer
Invasive cancer

6.4 After surgery

Radioactive iodine (I-131) ablates the thyroid remnant and treats microscopic or metastatic disease in intermediate- and high-risk patients. The patient is prepared by thyroxine withdrawal or recombinant TSH and a low-iodine diet, and iodinated contrast is avoided beforehand. Its side effects are sialadenitis, xerostomia, taste change, nasolacrimal duct obstruction and second cancers.

TSH suppression with levothyroxine aims for a TSH below 0.1 mU/L in high-risk disease and a low-normal level in low-risk disease. **Surveillance** follows thyroglobulin, with its antibodies, against the baseline, together with neck ultrasound, diagnostic radioiodine scans, and PET-CT for thyroglobulin-positive, iodine-negative disease. External beam radiotherapy treats unresectable or gross residual disease and tracheal invasion in older patients, and iodine-refractory progressive disease is treated with kinase inhibitors such as lenvatinib and sorafenib, and targeted agents for RET, NTRK and BRAF changes.

7 CHAPTER SEVEN Thyroidectomy

The operation and its complications.

7.1 Types

OPERATIONS ON THE THYROID

OPERATION	WHAT IS REMOVED	TYPICAL USE
Lobectomy (hemithyroidectomy)	One lobe with the isthmus	Benign nodule, follicular neoplasm, low-risk cancer, toxic adenoma
Isthmusectomy	The isthmus	Isthmic nodule; biopsy and airway relief in anaplastic carcinoma or lymphoma
Subtotal thyroidectomy	Both lobes leaving a small posterior remnant	Formerly Graves and multinodular goitre

OPERATION	WHAT IS REMOVED	TYPICAL USE
Near-total thyroidectomy	One lobe totally, the other leaving under 1 g near the nerve and parathyroid	Cancer, Graves, goitre
Total thyroidectomy	Both lobes and isthmus	Cancer, Graves, large multinodular goitre
Completion thyroidectomy	The remaining lobe after lobectomy	Lobectomy histology shows cancer needing total thyroidectomy

7.2 Indications and preparation

The indications are the four Cs plus one: cancer, suspicion of cancer, compression, cosmesis, and hyperthyroidism. Before operating, the patient should be euthyroid; calcium and vitamin D are measured, the larynx is examined, the thyroid and neck are scanned by ultrasound with FNA, CT is obtained for retrosternal extension, calcitonin is measured when medullary cancer is suspected, and the usual anaesthetic assessment is made.

7.3 Technique

1. **Incision:** a collar incision in a skin crease about two fingerbreadths above the sternal notch.
2. **Flaps:** subplatysmal flaps raised.
3. **Strap muscles:** separated in the midline and lifted off the gland.
4. **Lateral mobilisation:** the middle vein divided, and the gland rotated medially.
5. **Superior pole:** the vessels divided individually on the capsule, in the avascular cricothyroid (Reeve) space, to spare the external branch of the superior laryngeal nerve.
6. **The recurrent nerve:** identified and traced to its laryngeal entry. Nerve monitoring confirms and protects it, and loss of signal on the first side may stop a planned bilateral operation.
7. **Parathyroids:** each gland identified and kept on its blood supply by dividing inferior thyroid artery branches on the capsule; a devascularised gland is confirmed by frozen section, minced and reimplanted into sternomastoid.

8. **Berry's ligament:** divided carefully; cautery near the nerve is dangerous here.
9. **Closure:** haemostasis checked with a Valsalva manoeuvre, then straps and platysma closed. Drains are optional; they do not prevent haematoma.
10. **Minimal-access approaches:** video-assisted and remote-access robotic routes for selected patients.

OPERATIVE GUIDES IN THEATRE 3 operations

Open, video-assisted and robotic thyroidectomy.

Open Thyroidectomy
Thyroidectomy
Minimally Invasive Video-Assisted Thyroidectomy
Video-assisted
Robotic-Assisted Thyroidectomy
Robotic

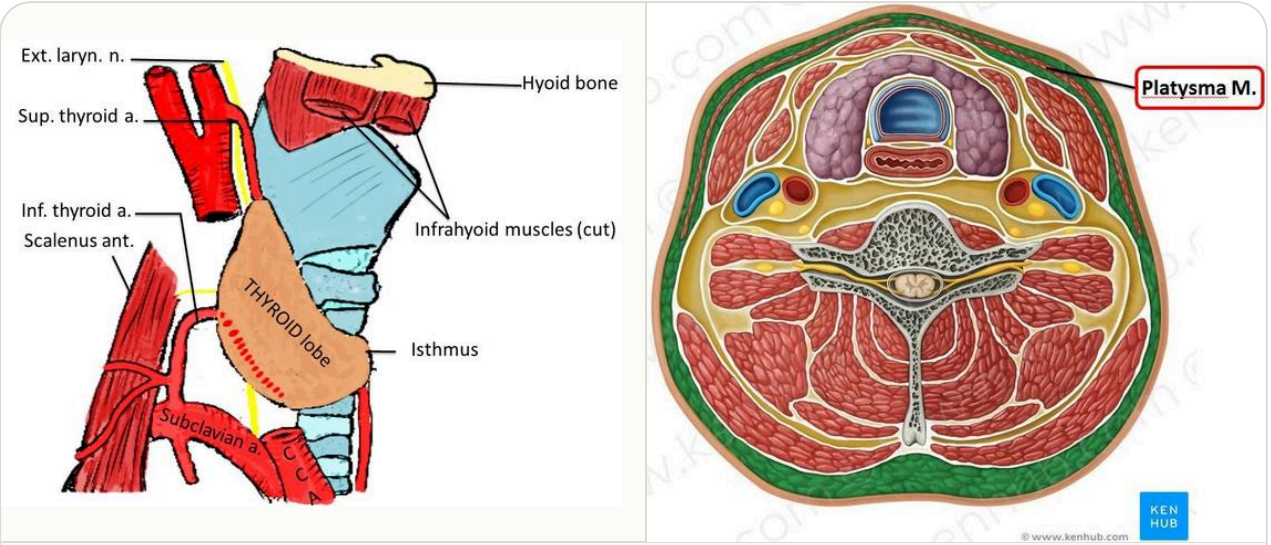


FIGURE 9 The thyroid with the strap muscles reflected, and a cross-section of the neck at the level of the thyroid.

Anatomy atlas · Doss p. 127, ENT Steps p. 273

7.4 Complications

Haematoma complicates under 2% of thyroidectomies, usually within hours, and is the complication that kills if it is not acted on at once.

NECK HAEMATOMA AFTER THYROIDECTOMY

Swelling, tight neck, stridor or distress:

1. Open the skin and the strap muscles at the bedside.

2. Evacuate the clot.
3. Secure the airway.
4. Return to theatre.

Venous and lymphatic congestion makes the larynx oedematous, so intubation is difficult. Release of pressure comes first.

Recurrent laryngeal nerve injury is temporary in up to about 5 to 10% and permanent in about 1%, with a higher risk in reoperation, cancer and retrosternal goitre. Unilateral injury causes hoarseness and aspiration. *Bilateral* injury may leave a near-normal voice but causes stridor on extubation, often needing reintubation and tracheostomy. Injury to the **external branch of the superior laryngeal nerve** costs high pitch and vocal projection, which matters greatly to singers.

Hypocalcaemia from hypoparathyroidism is common and transient after total thyroidectomy and permanent in about 1 to 3% in expert hands, more after central dissection. It appears at 24 to 72 hours with perioral and fingertip tingling, cramps, Chvostek's and Trousseau's signs, tetany, laryngospasm and a long QT. Calcium and PTH are measured after surgery, and hypocalcaemia is treated with oral calcium and calcitriol, or intravenous calcium gluconate when severe. Thyroid storm, rare tracheomalacia, expected hypothyroidism after total thyroidectomy, wound infection, seroma, keloid, pneumothorax, and chyle leak after lateral dissection complete the list.

WHERE THE BOOKS DIFFER · SIZE CUT-OFFS AND LOBECTOMY

Dhingra	Lee
Microcarcinomas under 1.5 cm and intrathyroidal tumours without nodes may have lobectomy; follicular neoplasms over 4 cm in the elderly go straight to total thyroidectomy; no elective neck dissection.	Following the 2015 ATA guidelines, lobectomy suffices for low-risk intrathyroidal tumours up to 4 cm; total thyroidectomy for larger, extrathyroidal or metastatic disease or when radioiodine is planned; prophylactic central dissection is debated.

Current practice follows the ATA approach and uses risk stratification rather than size alone.

BRITISH PRACTICE: U AND THY GRADES, NIFTP AND LOW-DOSE IODINE (OXFORD TEXTBOOK)

Thyroid cancer incidence has risen sharply over three decades while mortality has risen much less, a sign that imaging is finding many indolent cancers. British practice grades nodules on ultrasound from **U1** (normal) and **U2** (benign) to **U3** (indeterminate), **U4** (suspicious) and **U5** (malignant), and requires ultrasound-guided FNA for U3 to U5. Cytology is graded **Thy1** (non-diagnostic), **Thy2** (non-neoplastic), **Thy3a** (atypia), **Thy3f** (follicular neoplasm), **Thy4** (suspicious) and **Thy5** (malignant), roughly equivalent to the Bethesda categories. Thy3 and above, and any discordance between imaging and cytology, go to the thyroid multidisciplinary team. Thy3f and Thy4 lesions usually have a **diagnostic hemithyroidectomy**.

New entities include tumours of uncertain malignant potential and **NIFTP** (non-invasive follicular thyroid neoplasm with papillary-like nuclear features), which is now regarded as benign and needs no further treatment after lobectomy. **Completion thyroidectomy** is advised for tumours over 4 cm, multifocal or bilateral disease, extrathyroidal spread, familial disease, nodal or distant metastases. A solitary **papillary microcarcinoma** under 1 cm without adverse features needs only hemithyroidectomy without TSH suppression, and active surveillance is an option in selected patients.

Radioactive iodine is given for high-risk disease (tumour over 4 cm, extrathyroidal extension, distant metastases) and can usually be omitted in low-risk disease. Recombinant human TSH has largely replaced thyroid hormone withdrawal before treatment, and the **HiLo** and **ESTIMABL** trials showed that a low ablation dose (1.1 GBq) is as effective as a high dose (3.7 GBq) for low- and intermediate-risk patients. At 9 to 12 months, **dynamic risk stratification** using thyroglobulin, ultrasound and imaging re-grades each patient's risk and guides how hard TSH is suppressed and how closely they are followed.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Surgical anatomy:** Berry's ligament; the course of the RLN on each side, and the non-recurrent nerve; the external branch of the superior laryngeal nerve; the parathyroids and their blood supply.

- **Lingual thyroid:** scan before excision.
- **Work-up of a nodule:** TSH, ultrasound pattern, FNA, Bethesda categories; why FNA cannot diagnose follicular carcinoma.
- **The five cancers:** their spread, markers, MEN syndromes, RET testing, and phaeochromocytoma first.
- **Extent of surgery,** radioiodine, TSH suppression and thyroglobulin follow-up.
- **Complications:** haematoma (open at the bedside), nerve palsies, hypocalcaemia.

8.2 Common pitfalls

COMMON PITFALLS

- Operating for medullary carcinoma without excluding phaeochromocytoma.
- Calling a cystic lateral neck node in a young adult a branchial cyst without considering metastatic papillary carcinoma.
- Excising a lingual thyroid without checking for other thyroid tissue.
- Giving iodinated contrast just before planned radioiodine.
- Failing to check the cords before and after surgery.

Written for this app from Oxford Textbook of Otolaryngology ch. 59, Oxford Specialist Handbook ch. 11, Oxford Handbook of ENT ch. 15, Dhingra ch. 66 and K. J. Lee ch. 38, 51. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

THYROID & PARATHYROID · TUMOURS

Thyroid cancer

IMAGING FINDINGS

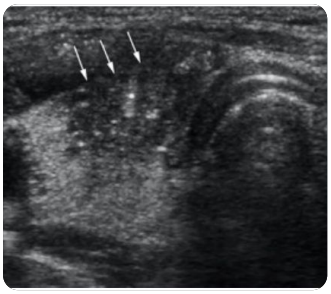
- Solid hypoechoic nodule with microcalcifications, taller than wide
- Irregular margins or extrathyroidal extension
- Cystic or calcified nodes in levels III, IV and VI

TOP DIFFERENTIALS

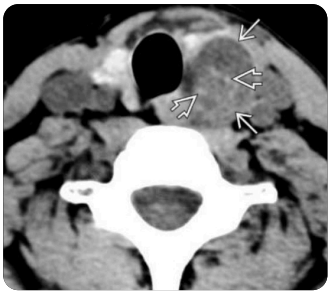
- Benign adenoma or colloid nodule
- Multinodular goitre
- Medullary or anaplastic carcinoma

CHECKLIST

- Trachea, oesophagus, larynx and recurrent nerve
- Iodinated contrast delays radioactive iodine



16-year-old lady presenting with hard right neck lump confirmed to be papillary ca on cytology. US shows strongly hypoechoic lesion with…



Axial NECT shows a well-defined mass arising from the left thyroid lobe with fine, speckled microcalcifications centrally. Small calcifications such as…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 238.

THYROID & PARATHYROID · INFECTION & INFLAMMATION

Hypertrophy of the Thyroid Gland, Thyroid Goiter

IMAGING FINDINGS

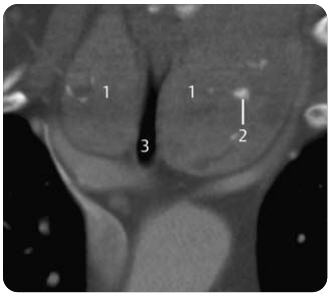
- Enlarged, heterogeneous gland with nodules, cysts and coarse calcification
- Tracheal deviation and narrowing
- Retrosternal extension

TOP DIFFERENTIALS

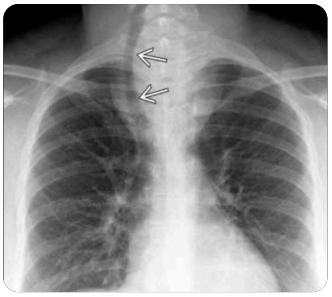
- Thyroid adenoma
- Differentiated or anaplastic carcinoma
- Lymphoma (rapid growth, Hashimoto)

CHECKLIST

- Lowest extent relative to the aortic arch and brachiocephalic vein
- Tracheal calibre for anaesthesia



CT with intravenous contrast enhancement, coronal. This slice at the level of the lower neck/upper thoracic cavity, shows enlarged thyroi…



Posteroanterior chest radiograph demonstrates marked displacement of cervical and thoracic trachea by a large, left-sided neck & superior…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 226.